

CS & IT ENGINEERING



Operating System

Disk Scheduling Algorithm

One Shot



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Recap of Previous Lecture



doubt?

Topic

File System

Topic

Blocks and File Allocation Methods



Topics to be Covered



Topic

Disk Scheduling

Topic

SSTF, Scan, Look



Topic : Question

[GATE-2022]



#Q. Consider two files systems A and B , that use contiguous allocation and linked allocation, respectively. A file of size 100 blocks is already stored in A and also in B. Now, consider inserting a new block in the middle of the file (between 50th and 51st block), whose data is already available in the memory. Assume that there are enough free blocks at the end of the file and that the file control blocks are already in memory. Let the number of disk accesses required to insert a block in the middle of the file in A and B are n_A and n_B respectively, then the value of $n_A + n_B$ is 153?

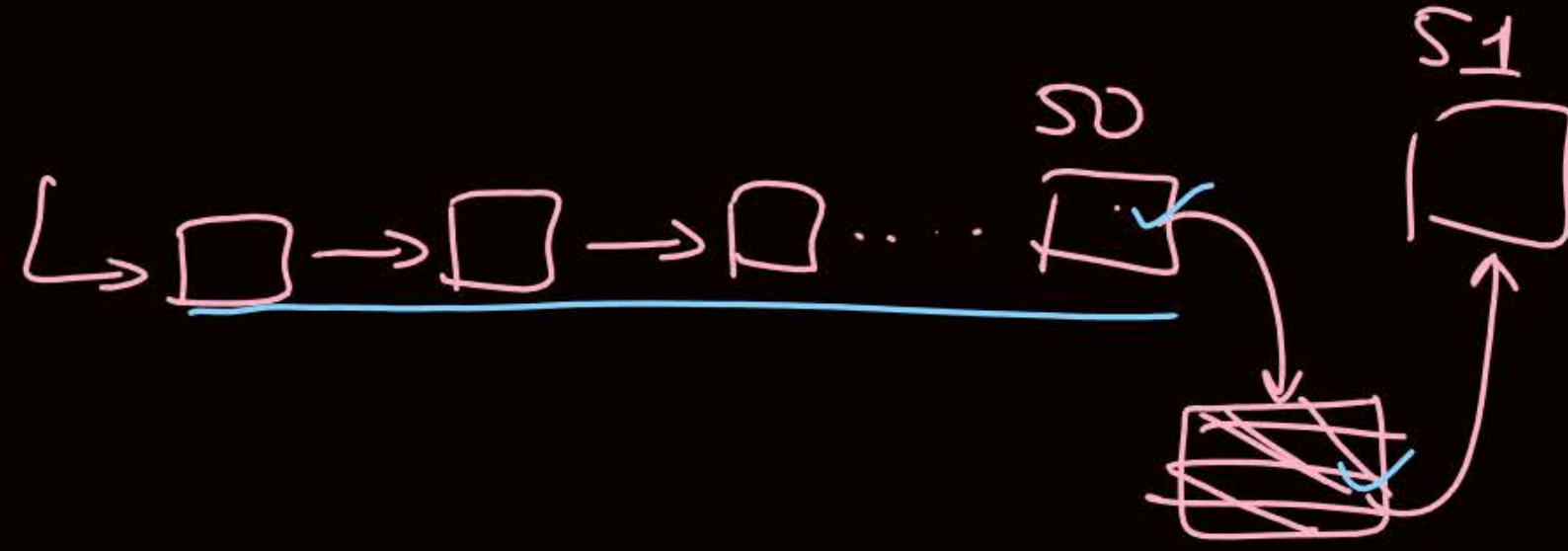
A
Conti
✓ 100 blocks

B
linked
✓

A:-

$$\begin{array}{rcl} 50 \text{ blocks shifted} & \Rightarrow & 100 \\ 1 \text{ insertion} & \Rightarrow & 1 \\ \hline n_A & = & 101 \end{array}$$

B:-



$$n_B = 52$$

$$\begin{aligned} n_A + n_B &= 101 + 52 \\ &= 153 \end{aligned}$$



Topic : Question

[GATE-2005]



#Q. In a computer system, four files of size 11050 bytes, 4990 bytes, 5170 bytes and 12640 bytes need to be stored. For storing these files on disk, we can use either 100 byte disk blocks or 200 byte disk blocks (but can't mix block sizes). For each block used to store a file, 4 bytes of bookkeeping information also needs to be stored on the disk. Thus, the total space used to store a file is the sum of the space taken to store the file and the space taken to store the book keeping information for the blocks allocated for storing the file. A disk block can store either bookkeeping information for a file or data from a file, but not both.

What is the total space required for storing the files using 100 byte disk blocks and 200 byte disk blocks respectively?



35400 and 35800 bytes



35800 and 35400 bytes



35600 and 35400 bytes



35400 and 35600 bytes



Topic : Solution

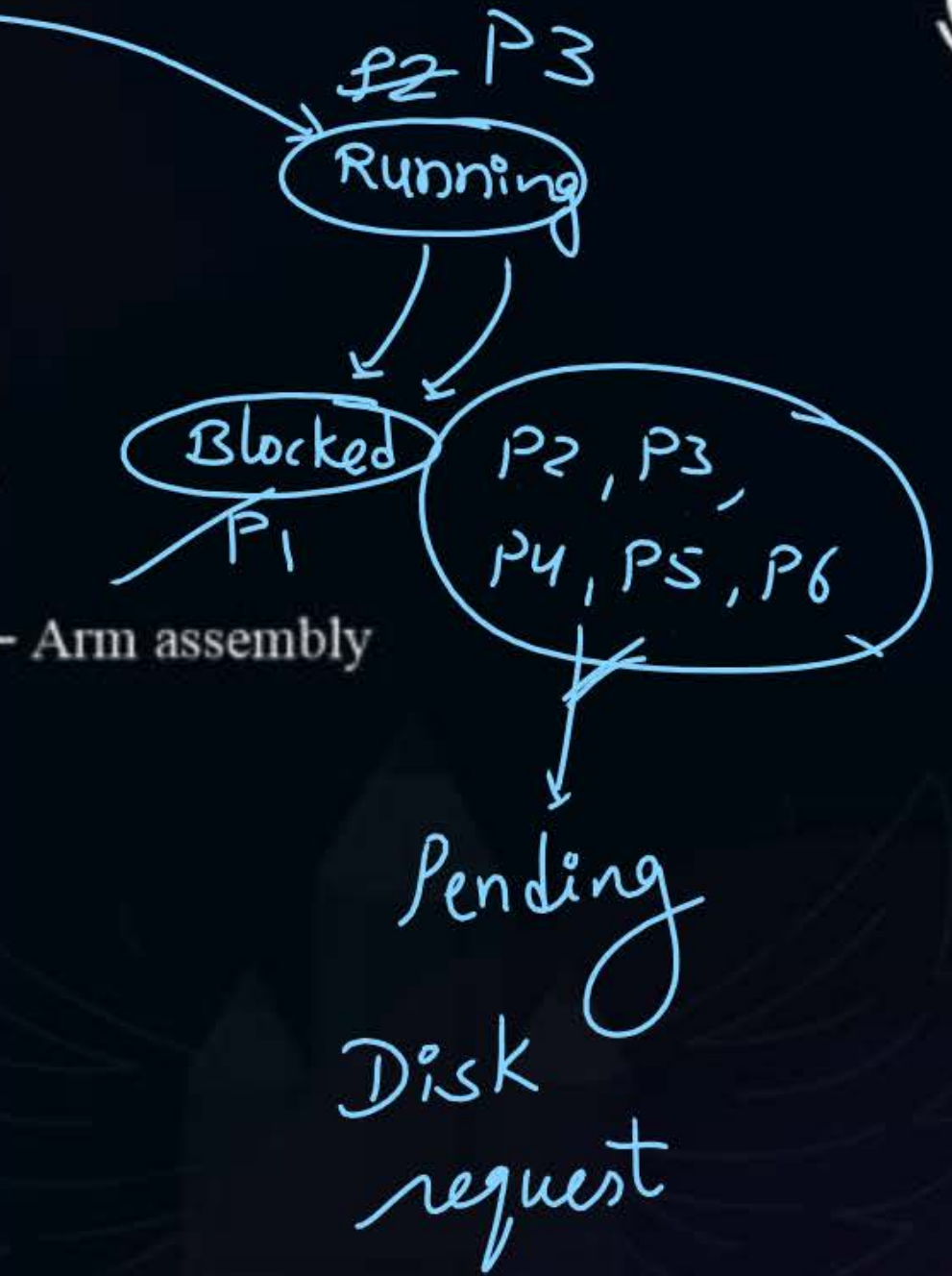
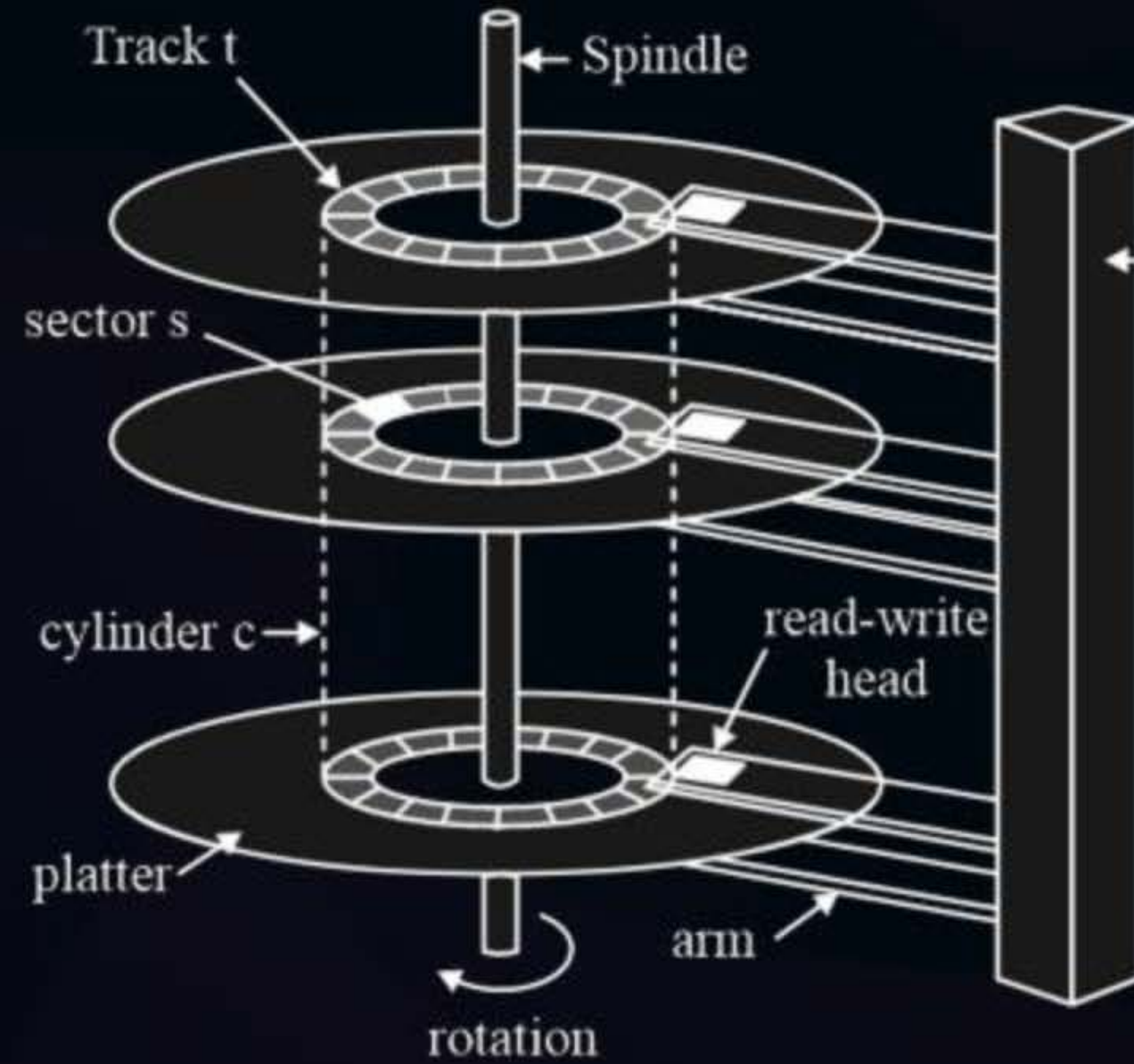
File Size	No. of blocks to store file	No. of blocks to store bookkeeping	Total blocks	Total Size
11050 Bytes	$\left\lceil \frac{11050}{100} \right\rceil = 111$	$\left\lceil \frac{111 * 4B}{100B} \right\rceil = 5$	116	
4990 Bytes	50	= 2	52	
5170 Bytes	52	= 3	55	
12640 Bytes	127	= 6	133	

356

$$356 * 100B = 35600B$$



Topic : Disk

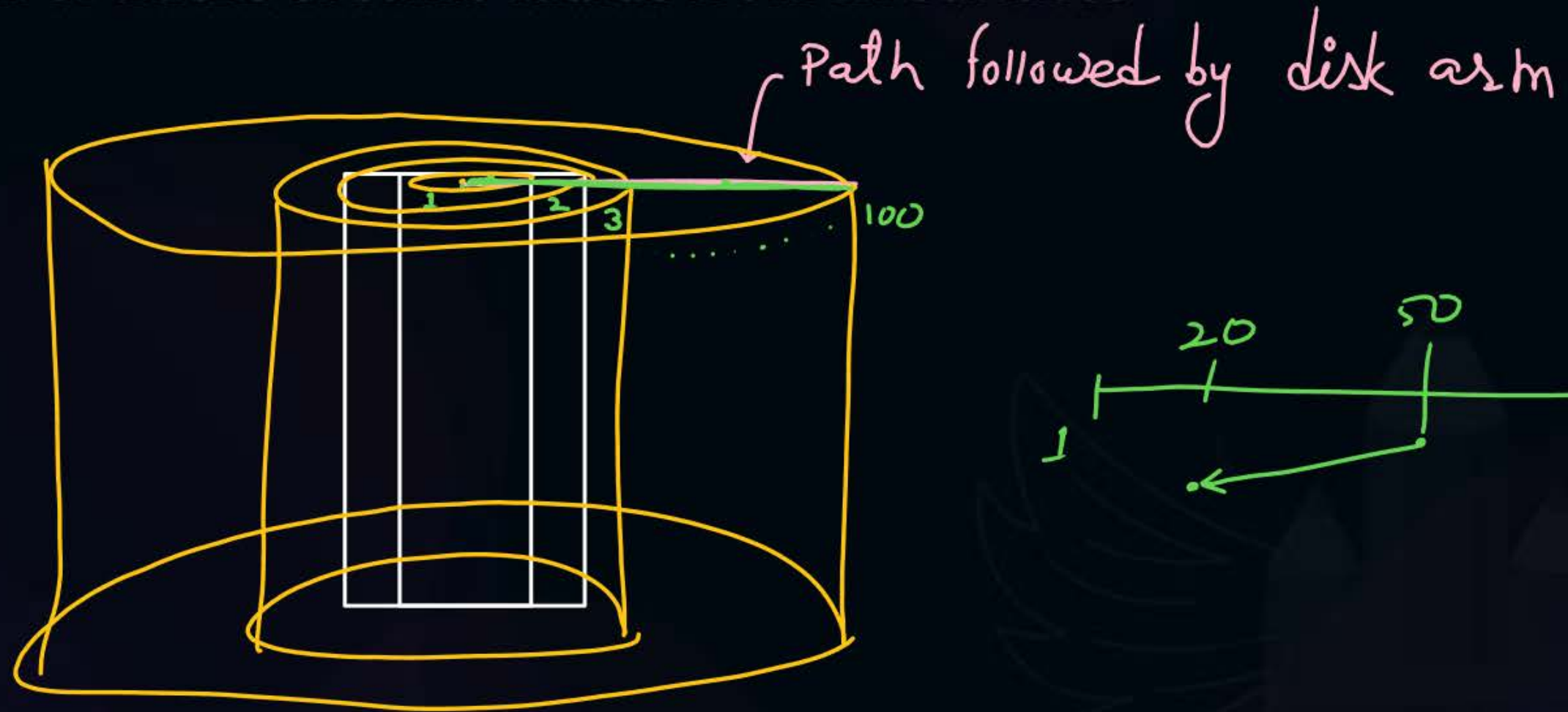




Topic : Cylinder



- Collection of tracks of same radius from all surfaces





Topic : Disk Scheduling

- Done by operating systems to schedule I/O requests arriving for the disk



Goal :- to move R/W head as min. as possible
to reduce seek time.





Topic : Disk Scheduling Algorithms

1. FCFS (First Come First Serve)
2. SSTF (Shortest Seek Time First)
3. Scan
4. C-Scan (Circular-Scan)
5. Look
6. C-Look (Circular-Look)



Topic : FCFS (First Come First Serve)

Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25



Topic : FCFS (First Come First Serve)

Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



no. of times
R/W head changed
direction = 4

$$\begin{aligned} \text{no. of cylinder movements} &= (36 - 25) + (80 - 36) + \\ &\quad (80 - 17) + (65 - 17) + \\ &\quad (65 - 7) + (7 - 3) + (90 - 3) \\ &= \underline{\underline{315}} \end{aligned}$$

in prev. ex. if each head movement takes 2ms, then

$$\begin{aligned}\text{total seek time} &= 315 * 2 \\ &= 630 \text{ ms}\end{aligned}$$

in prev. ex. if each head movement takes 3ms
then each direction change in head takes 10ms
then total seek time = 985 ?

$$\begin{aligned}&= (315 * 3) + (4 * 10) \\ &= 945 + 40 = 985 \text{ ms}\end{aligned}$$



Topic : FCFS (First Come First Serve)

Advantages:

- Every request gets a fair chance
- No indefinite postponement
 ↳ starvation

Disadvantages:

- Does not try to optimize seek time
- May not provide the best possible service



Topic : SSTF (Shortest Seek Time First)

↳ services nearest request first

Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25





Topic : SSTF (Shortest Seek Time First)

Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{Total head movements} &= 22 + 87 \\ &= 109\end{aligned}$$



Topic : SSTF (Shortest Seek Time First)

Advantages:

- Average Response Time decreases
- Throughput increases

Disadvantages:

- Overhead to calculate seek time in advance
- Can cause Starvation for a request if it has higher seek time as compared to incoming requests
- High variance of response time as SSTF favors only some requests



Topic : Scan (Elevator)



Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

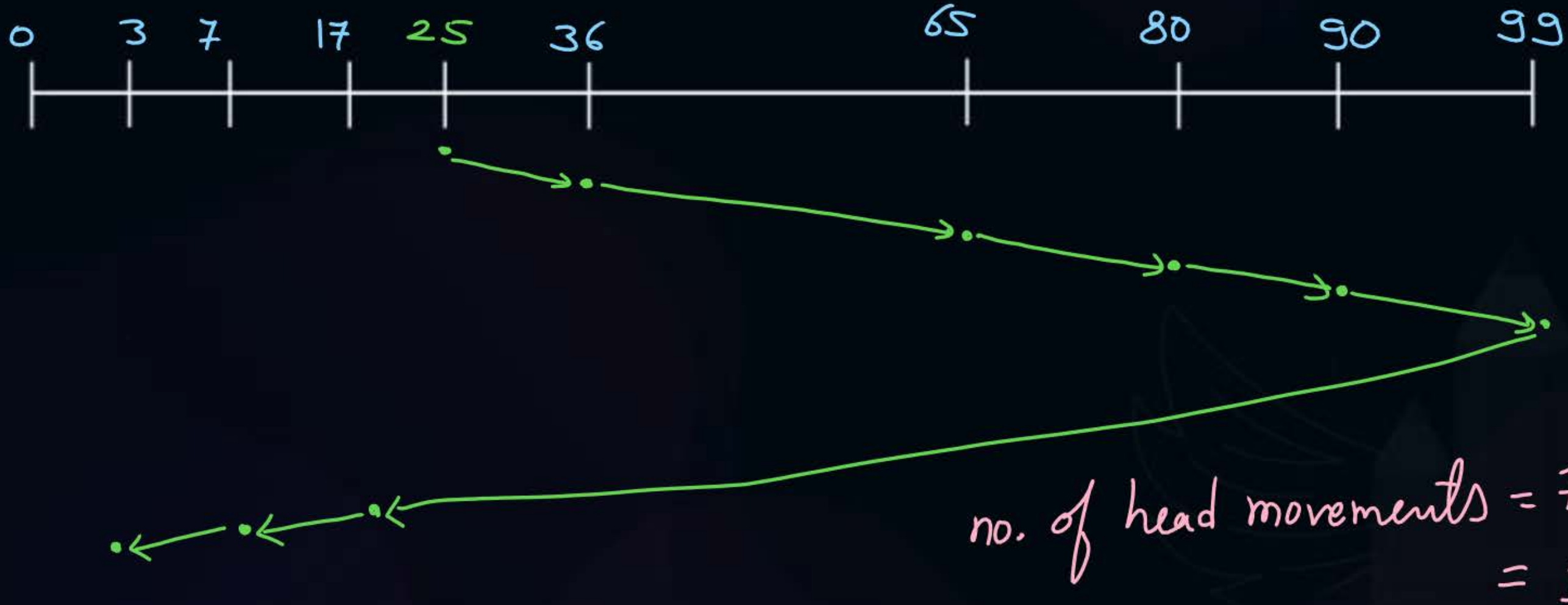
The arm should move “towards the larger value”

Direction will change either at 0 or at 99.



Topic : Scan (Elevator)

Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90





Topic : Scan



Advantages:

- High throughput
- Low variance of response time
- Average response time

Disadvantages:

- Long waiting time for requests for locations just visited by disk arm



Topic : C-Scan

(Circular - Scan)



Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

The arm should move “towards the larger value”

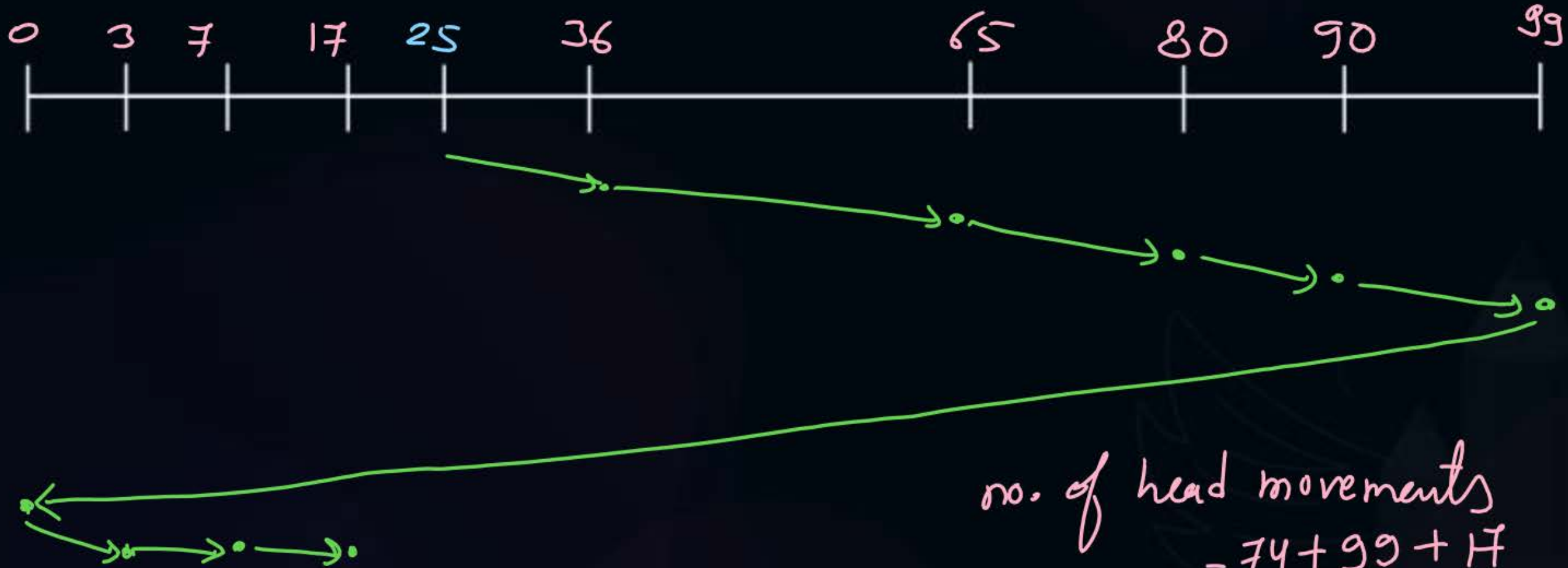




Topic : C-Scan



Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{no. of head movements} &= 74 + 99 + 17 \\ &= 190\end{aligned}$$



Topic : C-Scan



Advantages:

- Provides more uniform wait time compared to SCAN





Topic : Look



Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

The arm should move “towards the larger value”





Topic : Look



Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{no. of head movements} &= 65 + 87 \\ &= 152\end{aligned}$$



Topic : C-Look



Suppose total 100 cylinders 0-99

The order of request is: 36, 80, 17, 65, 7, 3, 90

The Read/Write arm is at 25

The arm should move “towards the larger value”

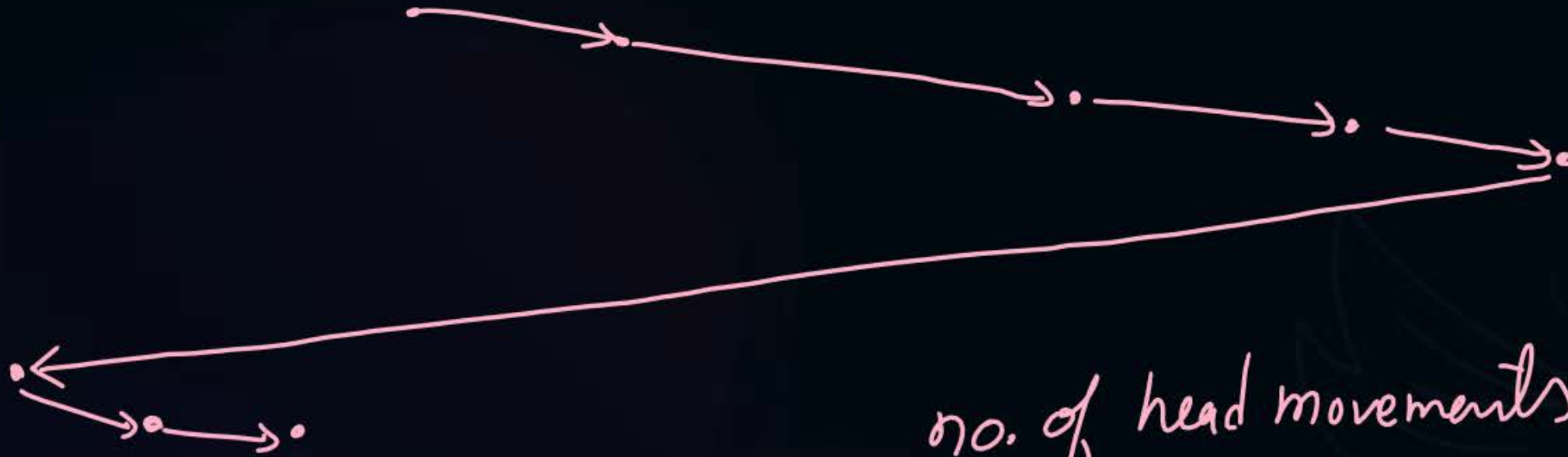




Topic : C-Look



Suppose the order of request is: 36, 80, 17, 65, 7, 3, 90



$$\begin{aligned}\text{no. of head movements} &= 65 + 87 + 14 \\ &= 166\end{aligned}$$

[MCQ]



uniprogramming

#Q. Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to give 50% better benchmark results, what is the expected improvement in the I/O performance of user programs? **[2004]**

A

50%

B

40%

C

25%

D

0%

#Q. The head of a hard disk serves request following the shortest seek time first (SSTF) policy. The head is initially positioned at track number 180.

Which of the request sets will cause the head to change its direction after servicing every request assuming that the head does not change direction if there is a tie in SSTF and all the request arrive before the servicing starts?

A ✗ 11, 139, 170, 178, 181, 184, 201, 265

B ✓✓ 10, 138, 170, 178, 181, 185, 201, 265

C ✗ 10, 139, 169, 178, 181, 184, 201, 265

D ✗ 10, 138, 170, 178, 181, 185, 200, 265

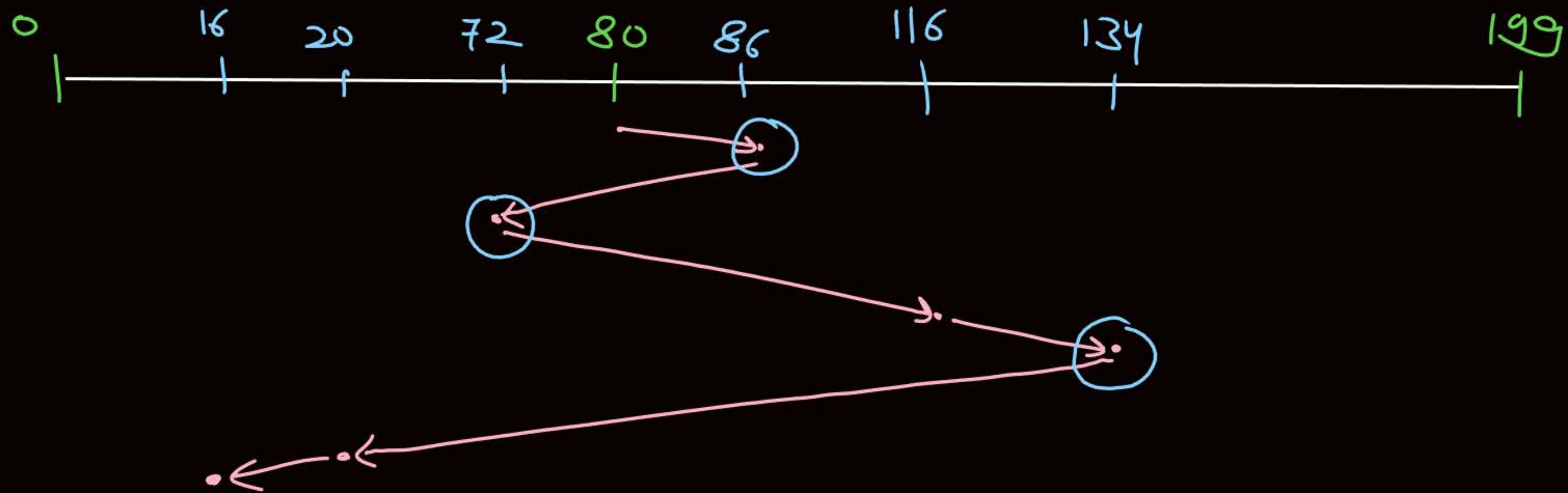
[2007]

#Q. Consider a storage disk with 4 platters (numbered as 0, 1, 2 and 3), 200 cylinders (numbered as 0, 1, ... , 199), and 256 sectors per track (numbered as 0, 1, ... 255). The following 6 disk requests of the form [sector number, cylinder number, platter number] are received by the disk controller at the same time:

[120, 72, 2], [180, 134, 1], [60, 20, 0], [212, 86, 3], [56, 116, 2], [118, 16, 1]

Currently head is positioned at sector number 100 of cylinder 80 and is moving towards higher cylinder numbers. The average power dissipation in moving the head over 100 cylinders is 20 milliwatts and for reversing the direction of the head movement once is 15 milliwatts. Power dissipation associated with rotational latency and switching of head between different platters is negligible. The total power consumption in milliwatts to satisfy all of the above disk requests using the Shortest Seek Time First disk scheduling algorithm is 85 . ?

[2018]



$$\begin{aligned} \text{Total head movements} &= 6 + 14 + 62 + 118 \\ &= 200 \end{aligned}$$

$$\text{Total power consumption} = \left(\frac{200}{100} * 20 \right) + (15 * 3) = 85 \text{ mW}$$



2 mins Summary

Topic

Disk Scheduling

Topic

SSTF, Scan, Look



Happy Learning

THANK - YOU

